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PI Name:	Mason, Christopher Ph.D.		
Project Title:	The Landscape of DNA and RNA Methylation Before, During, and After Human Space Travel		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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COI Name (Institution):	Melnick, Ari M.D. (Weill Medical College of Cornell University) Grills, George Ph.D. (Cornell University) Garrett-Bakelman, Francine M.D., Ph.D. (Weill Cornell Medicine)		
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Task Description:	Next-generation sequencing (NGS) technologies have recently been used to detect and classify epigenetic changes at an unprecedented level. Epigenetic patterns help define cellular phenotypes and can even be used to more accurately sub-classify human tumors such as acute myeloid leukemias (AML). Indeed, since some phenotypes such as cancer and aging can be better differentiated with epigenetic signatures (vs. purely genetic), epigenetic profiling is the focus of large-scale NGS efforts at the National Institutes of Health (NIH) like the Epigenomics Roadmap. There is also evidence that epigenetic changes accompany DNA damage and that sites of methylation correspond to DNA damage sites. Finally, there is an emerging field of "epitranscriptomics," where modifications of RNA, like the epigenetic modification of DNA, have been shown to impact brain development, gene regulation, and protein translation rates. Such ongoing research in DNA and RNA methylation has shown that these mediators of gene expression and modification represent critical regulatory layers in cell biology and genetics. Yet, these key epigenetic mechanisms of cellular regulation have barely, or never, been studied in the context of human space travel or during various gravity transitions ("G-transitions"). While previous work has addressed gene expression and proteomic dynamics during space travel, this Human Exploration Research Opportunities (HERO) solicitation from NASA regarding the Kelly twins provides an extraordinary opportunity to finally examine the impact of space travel on epigenetic, epitranscriptomic, and expression changes for identical twins. We propose to generate genome-wide maps of DNA and RNA methylation sites for the twins, as well a complete catalog of coding and noncoding RNA expression, to generate a 13 time-point, integrated portrait of gene regulation before, during, and after space travel. We will use established biochemical methods and open-source algorithms to profile these samples, which will reveal both the inter-twin differences and intra-person temporal changes in DNA and RNA. This thorough, genome-wide molecular portrait of the epigenetic, epitranscriptomic, and gene regulatory landscape of these twins will help address the potential epigenetic and long-term genome regulatory risk of space travel, and also discern if there is similarity of the twins' epigenetic changes to other phenotypes like aging and cancer. Specifically, we will create: (Aim 1) – Genome-wide epigenetic profiles of DNA methylation changes. We will use enhanced reduced representation bisulfite sequencing (eRRBS) to discover the precise sites of differentially methylated cytosines (DMCs) and differentially methylated regions (DMRs) for these twins. We will also estimate the number of cells (the clonality) that harbor methylation changes using our methylkit, eDMR, and MethClone algorithms. (Aim 2) – A comprehensive catalog of coding and noncoding, small and large RNAs. We will use total RNA-sequencing (RNA-seq) with ribo-depletion of the RNA samples to characterize the coding RNA and the non-coding RNAs, as well as micro-RNA (miRNA) sequencing of the small RNAs. We will then examine the impact of the DMCs and DMRs from Aim #1 on gene expression, alternative splicing of expressed genes, fusion events, and catalog the number of non-coding RNAs active in the genome during space travel. (Aim 3) – Transcriptome-wide maps of RNA methylation sites. We will use our published Methylated RNA ImmunoPrecipitation sequencing (MeRIP-seq) methods and algorithms (MeRIPPer, r-make) to examine the sites of RNA methylation that occur during space travel, and integrate the epitranscriptomic changes with those that appear in the gene expression and epigenetic levels. Having such a thorough molecular perspective will serve the long-term goals of the NASA program and also drive basic biological discoveries about the human genome's response to space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have used methclone to look at shifts in clonality of methylation and also published this work.
Task Progress:	OBJECTIVES: Next-generation sequencing (NGS) technologies have recently been used to detect and classify epigenetic changes in DNA. These epigenetic changes help differentiate cellular phenotypes, such as cancer and aging, and have therefore been the focus of research efforts by the National Institute of Health (NIH). Additionally, there is evidence that epigenetic changes and sites of methylation correspond to sites of DNA damage. Similar changes in DNA, RNA modifications have been shown to impact brain development, gene regulation, and protein translation rates. DNA and RNA methylation are critical regulatory components in cell biology and genetics. However, key epigenetic mechanisms of cellular regulation have not been studied in the context of human space travel or during gravity transitions. While previous work has addressed gene expression and proteomic dynamics during space travel, the data collected from identical twins provides a unique opportunity to examine the impact of space travel on epigenetic, epitranscriptomic, and expression changes between these two individuals. Using data from NASA's TWINS Study, this experiment generated: 1. Genome-wide epigenetic profiles of DNA methylation changes. 2. A comprehensive catalog of coding and noncoding, small and large RNAs. 3. Transcriptome-wide maps of RNA methylation sites. 4. Single molecule genome sequencing and shotgun sequencing to annotate expressed variants and epigenetic states. 5. Shotgun metagenome sequencing of buccal and oral swabs. APPROACH: To generate the genome-wide epigenetic profiles of DNA methylation changes, investigators used enhanced reduced representation bisulfite sequencing (eRRBS) to discover the precise sites of differentially methylated cytosines (DMCs) and differentially methylated regions (DMRs) for the twins. Additionally, investigators estimated the number of cells (the clonality) that harbor methylation changes using methylkit, eDMR, and MethClone algorithms. The comprehensive catalog of coding, noncoding, small and large RNAs was generated using total RNA-sequencing (RNA-seq) with ribo-depletion of the RNA samples to characterize the coding RNA and the noncoding RNAs, as well as micro-RNA (miRNA) sequencing of the small RNAs. Investigators also examined the impact of previously obtained DMCs and DMRs on gene expression, alternative splicing of expressed genes, and fusion events, and catalog the number of non-coding RNAs active in the genome during space travel. Investigators used published Methylated RNA ImmunoPrecipitation sequencing (MeRIP-seq) methods and algorithms (MeRIPPer, rmake) to examine the sites of RNA

	methylation that occur during space travel, and integrate the epitranscriptomic changes with those that appear in the gene expression and epigenetic levels. RESULTS: Methods for processing freshly collected pre-flight samples were determined and validated using flow cytometry for cell purity verification. Furthermore, downstream high molecular weight nucleic acid extractions for use were tested and validated for utilization in proposed studies. Preflight samples on both the inflight and ground subject were collected and processed.
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